

Rakesh Palani

Junior AI/ML Engineer | Generative AI | LLMs | RAG | Python

+91-9092733927 | rakeshpalani2004@gmail.com | Dharmapuri, Tamil Nadu, India

LinkedIn: linkedin.com/in/rakesh-palani-8bb1aa38b | GitHub: github.com/Rakesh051204 | Portfolio: stellar-glance-lab.lovable.app

Professional Summary

BCA graduate (CGPA 8.2/10) and AI/ML Engineer with hands-on experience independently building and deploying end-to-end Generative AI applications. Skilled in large language models (LLMs), retrieval-augmented generation (RAG), FastAPI, Python, SQL, embeddings, semantic search, vector databases, prompt engineering, multimodal ingestion, model routing, and LLM evaluation. Practical data science experience in exploratory data analysis, statistical analysis, machine learning, and Power BI.

Experience

Data Scientist Intern

Jul 2025 – Dec 2025

KGiSL MicroCollege

Python, SQL, Pandas, NumPy, scikit-learn, Power BI

- Performed data cleaning, exploratory data analysis, feature engineering, KNN imputation (k=4), correlation analysis, and linear regression on a 1,339-record, 44-feature coffee-quality dataset.
- Analyzed the relationship between altitude, acidity, and six coffee-quality attributes to support statistical interpretation and predictive modeling.
- Built an interactive Power BI dashboard using 50,000+ IMDb records to analyze rating, genre, and release-date trends for stakeholder review.

Projects

Stoic – AI Search and Answer Platform

2025 – Present

Python, FastAPI, RAG, LLMs, Supabase, pgvector, Groq

GitHub

- Designed and deployed an end-to-end RAG pipeline with semantic chunking, Voyage AI embeddings, HNSW cosine-similarity search, and Supabase pgvector for context-aware multi-turn AI search.
- Implemented live server-sent events (SSE) streaming, LLM-as-judge evaluation, multimodal ingestion for images, PDF, DOCX, and video, FFmpeg/Whisper processing, SearXNG search, and Groq model routing.

CodeZaro – AI Code Review Platform

2025 – Present

Python, FastAPI, REST APIs, LLM APIs, JWT, Fernet

GitHub

- Built and deployed a tiered AI code-review SaaS with Free, Pro, and Team plans, LLM-based static analysis using Groq Llama 3.3 70B, and configurable monthly usage limits.
- Implemented JWT access and refresh tokens, per-tier usage enforcement, and a FastAPI model-routing layer for Anthropic-to-Groq inference migration; secured BYOK key storage with Fernet encryption.

KinMax AI – Mock Interview Simulator

2025

Prompt Engineering, LLM APIs, React, Vite, Vercel

GitHub

- Created structured interview questions and LLM evaluation prompts to score candidate responses for technical clarity, accuracy, and communication; co-built and deployed the React/Vite application with a serverless API on Vercel.

Technical Skills

- **Generative AI:** LLMs, RAG, prompt engineering, embeddings, semantic chunking, semantic search, vector databases, pgvector, HNSW, LLM-as-judge evaluation, multimodal AI, model routing
- **Backend and APIs:** Python, FastAPI, REST APIs, PostgreSQL, Supabase, server-sent events (SSE), JWT authentication, Fernet encryption
- **Machine Learning and Data:** SQL, Pandas, NumPy, scikit-learn, EDA, KNN imputation, correlation analysis, linear regression, feature engineering, model evaluation, Power BI
- **Tools and Deployment:** Groq, Anthropic, Voyage AI, Llama 3.3 70B, Whisper, FFmpeg, SearXNG, React, Vite, Docker, Git, GitHub, Vercel, Render

Education

Bachelor of Computer Applications (BCA)

Graduated 2025

Vysya College of Arts and Science, Salem – Affiliated to Periyar University

CGPA: 8.2/10

Certifications and Achievements

- Statistics Foundations – Meta, Coursera (Jul 2025); Fundamentals of Machine Learning – NVIDIA via Whizlabs, Coursera (Jul 2025)
- Solved 234+ LeetCode problems covering arrays, dynamic programming, hash tables, and graph algorithms.